

Hydroponic & Aeroponic Biomass Production Guidelines

1. Automated Nutrient Film Technique

Continuous flow nutrient channels support dwarf wheat, sweet potato, and leafy greens under dual-spectrum red/blue LED arrays operating at 350 micromoles/m²/s.

2. Atmosphere Composition & Enrichment

Greenhouse modules maintain carbon dioxide enrichment at 1200 ppm and 65 percent relative humidity, accelerating crop vegetative growth cycles by 38 percent compared to terrestrial baselines.

3. Biological Waste Recirculation

Composted inedible biomass feeds anaerobic digestors, yielding bio-available nitrogen and methane while spirulina algae vats capture secondary carbon dioxide.